



Python Conditional Logic

Data with Dylan | Cheat Sheet | Python Fundamentals #2

WHAT IS CONDITIONAL LOGIC?

Programs choose paths from True/False checks.

```
age = 19
if age >= 18:
    print("Adult")
```

Tip: Think of it as a decision tree - if a condition is met, run that branch.

COMPARISON OPERATORS

OP	MEANING	EXAMPLE	RESULT
==	Equal to	5 == 5	True
!=	Not equal to	"5" != 5	True
>	Greater than	10 > 3	True
<	Less than	4 < 2	False
>=	Greater/equal	18 >= 18	True
<=	Less/equal	15 <= 10	False

Use == to compare values. Use = to assign values to variables.

TYPE / FORMATTING WARNINGS

▲ Missing colon after the condition

```
if age >= 18
    print("Adult")
# X SyntaxError
```

```
if age >= 18:
    print("Adult")
# correct
```

▲ Indentation controls what belongs inside the if block

```
if score >= 90:
print("Great")
# X IndentationError
```

```
if score >= 90:
    print("Great")
# correct
```

▲ input() returns text - convert before numeric comparisons

```
age = input("Age: ")
if age >= 18:
# X TypeError
```

```
age = int(input("Age: "))
if age >= 18:
# correct
```

MINI PRACTICE

Ask for a number and print:

"Too low" if < 10 | "Perfect" if == 10 | "Too high" if > 10

```
num = int(input("Enter a number: "))
```

Hint: use if / elif / else, ==, colon (:), and indentation

KEY SYNTAX & OPERATIONS

Basic if statement:

```
score = 95
if score >= 90:
    print("Great job!")
```

if / elif / else chain:

```
num = 7
if num < 0:
    print("Invalid")
elif num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

User input + type conversion:

```
age = int(input("Enter your age: "))
if age >= 18:
    print("You can vote")
```

LOGICAL OPERATORS

OP	USE WHEN...	EXAMPLE
and	Both checks must be True	age >= 18 and has_id
or	At least one check is True	is_student or has_coupon
not	Reverse the condition	not logged_in

```
age = 19
has_id = True
if age >= 18 and has_id:
    print("Access granted")
```

COMMON BEGINNER MISTAKES

X Using = instead of ==

```
if password = "abc123":
    # X SyntaxError
✅ if password == "abc123":
```

X Separate if statements for one choice

```
if grade >= 90: print("A")
if grade >= 80: print("B")
✅ use elif so only one branch runs
```

X Comparing input text to a number

```
guess = input("Number: ")
if guess == 5:
    # X TypeError
✅ guess = int(input("Number: "))
```

X Adding a condition after else

```
else age >= 18:
    # X SyntaxError
✅ elif age >= 18:
```

Game idea: compare guess vs secret using ==, <, and else for high/low hints.